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A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption

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Abstract
Accurate forecasting of household water consumption is essential for sustainable water-resources planning; however, guiding model selection remains challenging due to fragmented and heterogeneous evidence. Following the PRISMA methodology, we screened 4264 records from major electronic databases, including ScienceDirect and Web of Science, and identified 80 primary studies published between 2009 and 2024. The evidence is systematically synthesized to characterize dominant machine-learning model families, predictor patterns, and evaluation practices, and to translate these findings into actionable guidance for model selection. The analysis shows that multilayer perceptrons and recurrent architectures (e.g., LSTM) are predominantly used for sequence-based forecasting tasks relying on historical consumption and climatic predictors, whereas tree-based ensemble models are more frequently adopted in heterogeneous tabular settings incorporating property-related descriptors. RMSE and MAE emerge as the most commonly reported evaluation metrics across studies. Finally, the review highlights key research gaps and future directions, including the limited use of socio-economic predictors, the need for standardized hybrid modeling pipelines, and the importance of improving model interpretability for operational deployment.

Keywords Machine learning · Predictors · Water consumption forecasting · Systematic literature review

1 Introduction

Water scarcity is increasingly recognized as a global challenge, driven primarily by human-induced factors such as population growth, climate change, and industrial activities (Liu et al. 2017; Tzanakakis et al. 2020). As a result, water stress is now experienced across extensive regions, and the importance of sustainable planning and demand-oriented management strategies has been widely acknowledged. Although agricultural activities account for the largest share of global water withdrawals, household water use represents a significant and non-negligible component of total demand (Rafiei Sardooi et al. 2023; Siyal et al.

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2023). Global household water consumption levels were estimated to range from approximately 125 to over 600 liters per capita per day in 2020, depending on living standards and regional conditions (Seelen et al. 2019; The Government of Chile 2015). Consequently, accurate forecasting of household water consumption has emerged as an important policy and planning instrument (Sattler et al. 2023; Shen and Varis 2000).

In recent years, the water-resources literature has increasingly adopted data-driven and machine-learning (ML) approaches to support forecasting and decision-making under climatic and operational uncertainty. Recent studies have demonstrated the effectiveness of ML models across a broad range of water-management applications, including rainfall forecasting, flood prediction, and reservoir water-level estimation, often reporting improved predictive performance compared to traditional statistical or conceptual models (Akinsoji et al. 2024; Kumar et al. 2024; Rehamnia and Mahdavi-Meymand 2025; Schmitz et al. 2025). These contributions reflect a broader methodological shift toward data-driven modeling frameworks in water-resources research, particularly for systems characterized by nonlinear dynamics, heterogeneous predictors, and complex temporal dependencies.

At the household scale, water-consumption forecasting is commonly addressed using ML models trained on observations recorded by smart water meters and other end-device monitoring systems (Bibri 2018; Kim 2020). A growing body of studies has applied machine learning techniques to household-level demand forecasting for operational planning and demand management (Atitallah et al. 2020; Zubaidi et al. 2020). These efforts are distinct from water-quality monitoring, service-level assessment, or infrastructure-oriented studies such as leakage detection and network optimization (Islam et al. 2024; Moeinzadeh et al. 2024). Despite increasing interest, the literature remains fragmented, with a wide range of traditional statistical models, ensemble methods, and deep learning architectures being applied under heterogeneous data conditions.

Moreover, existing studies frequently report limited or inconsistent evidence regarding the systematic superiority of highly complex models over simpler alternatives, particularly when data availability, temporal resolution, and predictor selection vary substantially across contexts (Bischl et al. 2025; Lumbreras and Ciller 2025). As a result, model selection for household water-consumption forecasting depends not only on predictive accuracy, but also on data granularity, predictor availability, computational requirements, and deployment constraints at the household or end-device level (Cina et al. 2025; Sarıman and Keçebaş 2026).

Against this backdrop, a systematic synthesis of machine-learning approaches for household water-consumption forecasting is critically needed. This study presents a Systematic Literature Review (SLR) conducted in accordance with the PRISMA-2020 guidelines (Page et al. 2021), focusing exclusively on ML-based household-level consumption forecasting. Unlike studies operating at the system or infrastructure scale, this review concentrates on end-device and household-level prediction. A total of 4264 records were screened, resulting in 80 primary studies included for detailed analysis. The contributions of this review are fourfold: (i) it provides a comprehensive PRISMA-based synthesis of ML studies targeting household water-consumption forecasting; (ii) it introduces a structured taxonomy of models, predictors, and evaluation metrics; (iii) it develops a model-predictor crosswalk grounded in empirical evidence and operational considerations; and (iv) it characterizes the structural influence of key contributions through citation-network analysis.

The remainder of this paper is organized as follows. Section 2 describes the review methodology. Section 3 presents the descriptive statistics and the resulting taxonomy. Section 4 discusses the main findings, while Section 5 provides a practitioner-oriented decision framework. Section 6 situates the review within related work, Section 7 addresses threats to validity, and Section 8 outlines the conclusions and future research directions.

2 Methodology

This study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-2020) guidelines to ensure transparency, reproducibility, and methodological rigor (Page et al. 2021). PRISMA provides a structured framework for identifying, screening, and synthesizing evidence and is widely adopted in systematic reviews addressing machine-learning applications in water-resources research.

The review is guided by the following research questions (RQs), which define the scope and structure of the analysis:

- RQ1: What are the primary characteristics of machine learning techniques employed in predicting home water consumption?
- RQ2: What should take into consideration when choosing an ML technique according to home water consumption?
- RQ3: What performance metrics are arguably the best to evaluate machine learning models predicting household water consumption?
- RQ4: What are the challenges expected in this field?

A comprehensive literature search was conducted across multiple major digital libraries to ensure broad and interdisciplinary coverage of relevant studies. The databases consulted include ACM Digital Library, IEEE Xplore, Web of Science, ScienceDirect, Scopus, SpringerLink, and Wiley Online Library. These sources collectively provide extensive coverage of peer-reviewed research in computer science, engineering, environmental sciences, and water-resources management. Both journal articles and conference papers written in English were considered, reflecting the prominent role of conference venues in disseminating advances in machine-learning research.

To capture studies specifically addressing household or residential water-consumption forecasting using machine-learning techniques, the search strategy was structured around three thematic keyword groups: (i) machine-learning methods, (ii) household or residential water consumption, and (iii) forecasting or prediction tasks. Boolean operators were used to combine these keyword groups and refine the search across all databases. The complete search strings, keyword combinations, and query syntax for each digital library are reported in the Supplementary Material (https://doi.org/10.5281/zenodo.18131836).

The study selection followed a multi-stage screening process in accordance with PRISMA recommendations. An initial set of 4264 records was identified through database searching. After the removal of duplicates, titles and abstracts were screened for relevance with respect to the defined scope. Full-text screening was subsequently conducted using predefined inclusion and exclusion criteria. This process resulted in the selection of 80 primary studies for qualitative and quantitative synthesis. The overall identification, screening,

eligibility, and inclusion stages are summarized in the PRISMA flow diagram shown in Fig. 1, while detailed eligibility criteria are provided in the Supplementary Material (https://doi.org/10.5281/zenodo.18131836).

For each selected study, a structured data-extraction procedure was applied to collect information relevant to the research questions. Extracted attributes included machine-learning model families, predictor variables, temporal resolution, forecasting horizon, evaluation metrics, and application context. To maintain conciseness in the main manuscript, the complete data-extraction template, coding scheme, and taxonomy definitions are reported in the Supplementary Material.

The synthesis of results was performed using descriptive and comparative analysis techniques. Quantitative summaries were generated using counts, frequencies, and percentages to identify patterns and trends across studies. Visualizations were employed to support the synthesis of model usage, predictor categories, forecasting configurations, and evaluation practices. Additional methodological details, including coding decisions, cross-study aggregation procedures, and visualization workflows, are provided in the Supplementary Material.

To support this synthesis and enable systematic cross-study comparison, the extracted evidence was organized using a structured taxonomy of machine-learning model families, predictor categories, and evaluation-metric families. The taxonomy was developed by combining established domain conventions with patterns emerging from the reviewed studies. Prior work on forecasting models, predictor categorization, and validation practices informed category design (Adamowski and Chan 2015; Fu et al. 2022; Garzón et al. 2022; Gonzalez-Cebollada et al. 2020; Herath and Mittal 2022; Mosavi et al. 2018; Shen 2018; Sit et al. 2021; Zubaidi et al. 2020), while established guidelines for taxonomy construction

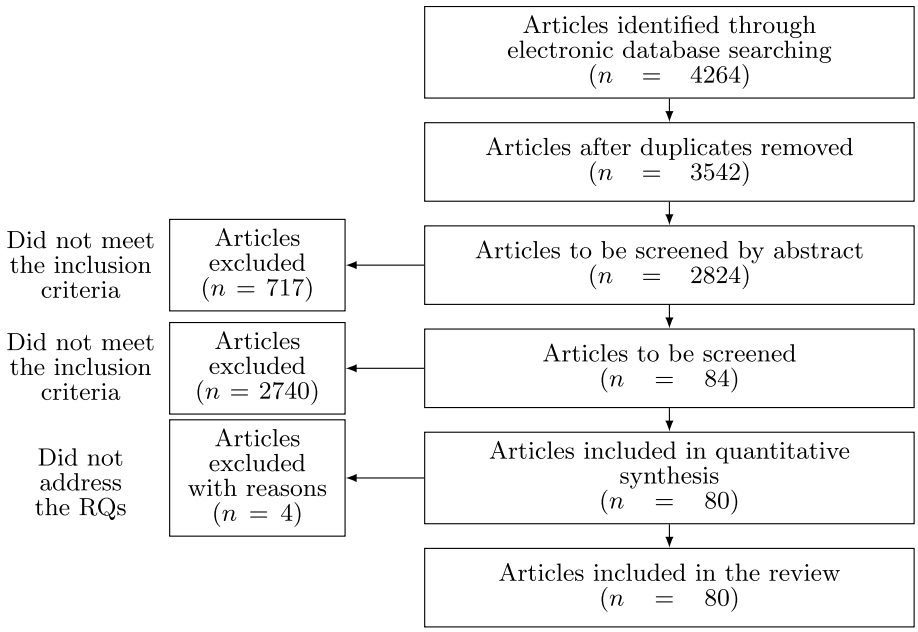

Fig. 1 PRISMA flow diagram of article selection

and evidence coding were followed to ensure conceptual consistency (Krippendorff 2018; Marzi et al. 2025; Nickerson et al. 2013). Detailed category definitions and coding rules are provided in the Supplementary Material.

3 Results

This section presents the main results of the systematic literature review on machine-learning approaches for forecasting household water consumption. The findings reported here are derived from the set of primary studies selected through the review protocol described in Section 2 and are organized to address the four research questions defined in this work.

The complete list of selected studies is provided in the Supplementary Material (DOI: 10.5281/zenodo.18131836), including a unique identifier for each study, full bibliographic information, and publication type. Among the selected studies, 59 (74%) were published in peer-reviewed scientific journals, while 21 (26%) appeared in conference proceedings. This distribution reflects established dissemination practices in the machine-learning community, where conference venues play a central role in reporting methodological advances, complemented by journal publications that emphasize applied and longitudinal studies.

The results are organized by first presenting an overview of the included studies, followed by a taxonomy-based synthesis addressing modeling approaches, predictor usage, and evaluation practices in relation to the defined research questions. For clarity, individual primary studies are referenced throughout the manuscript using identifiers (P01, P02, ..., P80), which correspond to the full bibliographic entries listed in the Supplementary Material.

3.1 Overview of Included Studies

Following the PRISMA-based selection process, a total of 80 primary studies were retained for analysis. These studies collectively represent the current state of research on machine-learning-based forecasting of household water consumption. The body of literature spans multiple publication years and reflects a growing interest in data-driven approaches for demand prediction at the household and end-device levels.

The selected studies were published across a diverse set of venues, including both journal articles and conference proceedings. This distribution reflects established publication practices in machine-learning and engineering research, where conference venues often serve as primary outlets for methodological advances, while journal publications tend to emphasize application-driven and longitudinal analyses.

From a methodological perspective, the reviewed studies exhibit substantial heterogeneity in terms of modeling approaches, predictor selection, temporal resolution, and evaluation practices. A wide range of machine-learning techniques has been applied, encompassing traditional statistical models, tree-based and ensemble methods, as well as deep learning architectures. Similarly, the predictors employed across studies vary considerably, including historical consumption records, meteorological variables, socio-demographic attributes, and contextual or calendar-based features.

Despite this diversity, common structural patterns emerge across the literature. In particular, historical consumption data appear as the most frequently used predictors, while advanced model architectures are not consistently associated with superior predictive per-

formance. These observations motivate the need for a structured taxonomy capable of organizing heterogeneous methodological choices and supporting cross-study comparison.

Detailed distributions of publication years, machine-learning techniques, and other descriptive statistics are provided in the Supplementary Material. The following subsection introduces the taxonomy developed to systematically synthesize the reviewed studies.

3.2 Taxonomy of Machine-learning Approaches for Household Water Consumption Forecasting

Figure 2 introduces a three-part taxonomy to standardize terminology and guide the synthesis of results for RQ1–RQ3, distinguishing among *(i)* machine-learning model families, *(ii)* predictor categories, and *(iii)* evaluation-metric families. This structure supports systematic comparison across heterogeneous studies by defining stable category boundaries and explicit inclusion rules.

Recent forecasting studies demonstrate substantial methodological heterogeneity in modeling strategies, feature design, and evaluation practices. For example, existing work

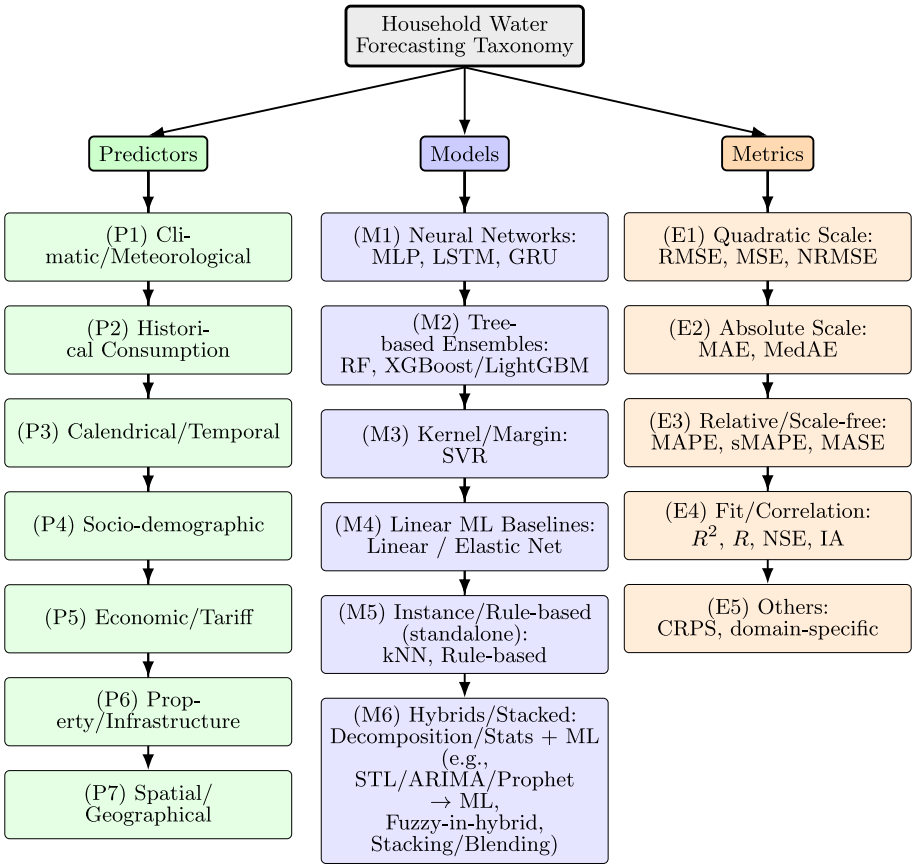

Fig. 2 Top-down taxonomy (evidence-aligned). M3 excludes Gaussian Processes; M4 lists only Linear/Elastic Net; ARIMA/Prophet and Fuzzy appear only within hybrids (M6)

contrasts neural architectures, ensemble-based approaches, and hybrid sequential models for short- and mid-term household water-consumption forecasting (Ghannam and Hussain 2024; Niknam et al. 2022). At the predictor level, inconsistent naming and aggregation of climatic, socio-demographic, infrastructural, and temporal variables have been widely reported, limiting cross-study comparability (Velasquez and Nguyen 2025; Yang et al. 2024). Evaluation practices are similarly fragmented, with metrics such as RMSE, MAE, MAPE, R^2, NSE, and normalized variants often used interchangeably, producing conflicting performance rankings and weakening external validity (Ahmed Osman et al. 2025). Formalizing the evidence into standardized model, predictor, and metric families directly addresses these issues and aligns with recent recommendations for taxonomy-driven synthesis frameworks in environmental machine-learning forecasting (Ahmed Osman et al. 2025; Yang et al. 2024).

Beyond the individual dimensions captured by the proposed taxonomy, cross-dimensional analysis reveals recurring patterns in how machine-learning models, predictor categories, and evaluation practices are combined in household water-consumption forecasting studies. This synthesis highlights practical trends and methodological dependencies that are not evident when dimensions are examined in isolation.

Across the reviewed studies, complex model architectures, including deep neural networks and hybrid sequential models, are most frequently paired with high-frequency historical consumption data and calendar-based predictors. In contrast, simpler statistical and tree-based models are more commonly applied when predictor sets are limited or when data are aggregated at coarser temporal resolutions. This pattern suggests that model complexity is often driven by data availability and granularity rather than by consistent evidence of superior predictive performance.

Evaluation practices further reflect this dependency structure. Studies employing advanced models frequently report multiple error metrics, including RMSE, MAE, and normalized variants, whereas simpler models are often evaluated using a narrower set of metrics. However, cross-study comparison remains challenging due to inconsistent metric selection and normalization procedures, particularly when forecasting horizons differ.

3.3 Primary Characteristics of the Machine Learning Techniques (RQ1)

To address RQ1, the key characteristics attributed to the machine learning (ML) techniques in residential water consumption forecasting were classified into 20 high-level conceptual categories. This categorization reduces semantic variability across the 80 studies while preserving the essential traits emphasized by the authors. Figure 3 illustrates the distribution of these IDs across categories, while Fig. 4 highlights their relative frequency.

The analysis reveals that Nonlinear modeling capability is the most prioritized feature, appearing in 28 out of 80 studies. This is followed by Reliability and the use of Improved versions of another model, highlighting a field-wide emphasis on model stability and iterative refinement over mere novelty.

Further insights are provided in Fig. 5, which presents a temporal bubble chart visualizing the co-evolution of techniques and their attributed characteristics. The visual flow reveals a significant "capability shift" around 2020. While earlier research (pre-2020) focused on older methods and basic benchmarking, the post-2020 period shows a rapid

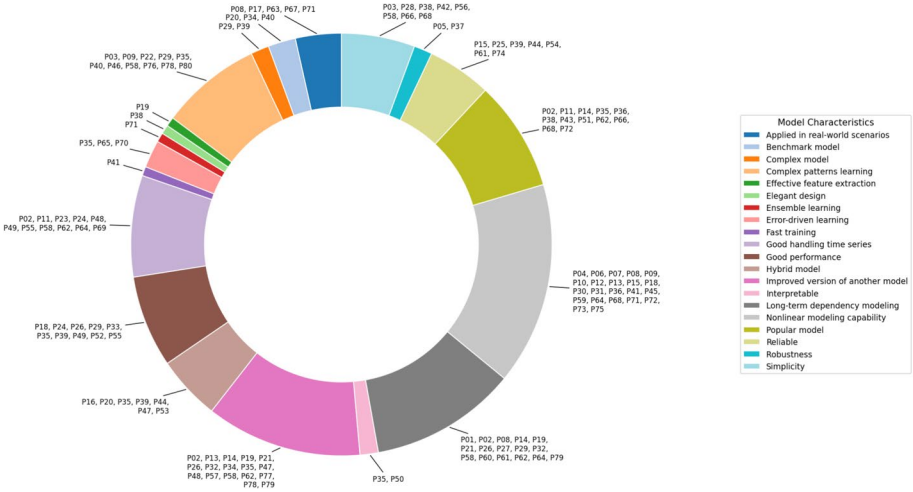

Fig. 3 Categorization of machine learning model characteristics identified in the reviewed studies

expansion of LSTM adoption alongside a surge in Long-term dependency modeling and Nonlinear modeling capability.

This visual correlation confirms that the community's move toward deep learning is functionally driven by the need to capture complex temporal dynamics that traditional models could not adequately address. Consequently, the answer to RQ1 suggests that researchers currently prioritize flexibility and temporal depth, reflecting the multifactorial nature of household water consumption.

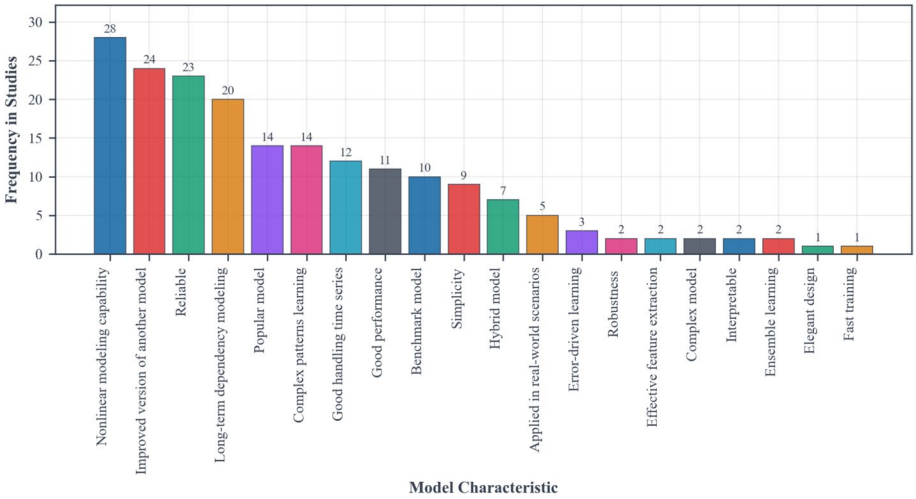

Fig. 4 Distribution of frequently cited model characteristics. Nonlinear modeling capability and reliability dominate, suggesting emphasis on flexibility and robustness in recent ML approaches

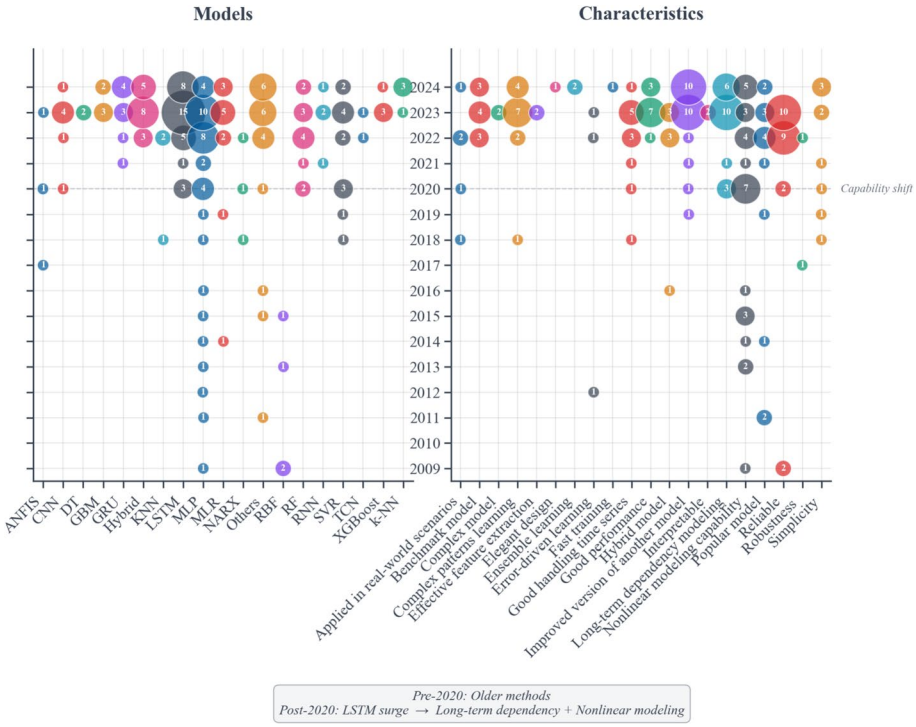

Fig. 5 Temporal bubble plot of ML models and characteristics. Two-part plot showing frequency of model usage and associated characteristics over time

3.4 Factors Used by Machine Learning Techniques for Home Water Consumption Prediction (RQ2)

Regarding RQ2, the variables employed for estimating residential water consumption were organized into eight thematic categories in order to facilitate synthesis across studies. The frequency of predictor usage is summarized in Fig. 6, while their temporal evolution across the 2009-2024 period is illustrated in Fig. 7.

Historical consumption variables were identified as the most frequently employed predictors, appearing in 47 of the 80 reviewed studies (58.75%). These variables are commonly used to capture recurrent usage patterns, seasonality, and short-term behavioral persistence. However, several studies (e.g., [P07], [P36]) reported that exclusive reliance on historical lags may constrain predictive accuracy, particularly in the presence of structural changes in consumption behavior or external perturbations. Climatic variables, including temperature and precipitation, were incorporated in 45 studies and represent the second most prevalent predictor category. Their increasing adoption after 2022 reflects growing attention to environmental drivers and climate-related variability in residential water demand.

Demographic predictors were employed in 35 studies, while socioeconomic factors were included in 26 studies. These variables were primarily associated with medium- and long-horizon forecasting tasks, where temporal information alone is insufficient to explain persistent differences in consumption across households. The inclusion of such predictors enables

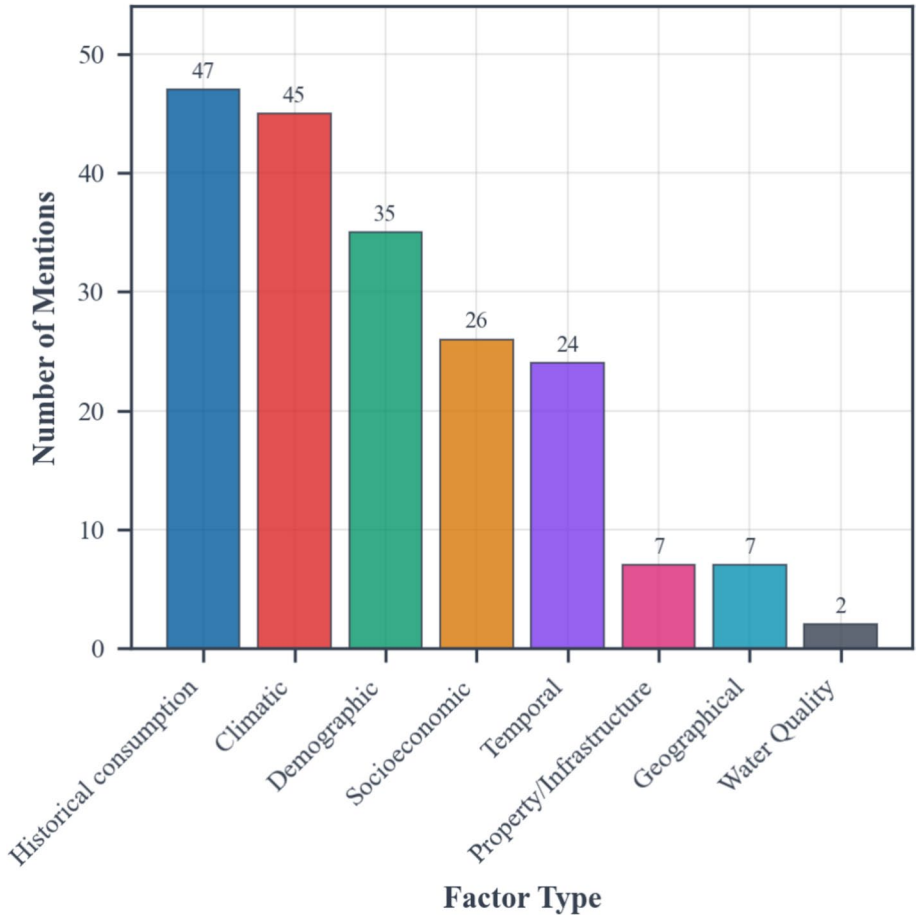

Fig. 6 Frequency of mentions for different factor types in household water consumption prediction models. Climatic and historical data were the most common, while geographical and water quality factors were underutilized

partial representation of population heterogeneity and demand elasticity. Nevertheless, their integration remains inconsistent across the literature, largely due to limitations in data availability and concerns related to privacy and regulatory compliance.

Property and infrastructure attributes, geographical variables, and water quality indicators were observed to be sparsely utilized throughout the reviewed period. As illustrated in Fig. 7, these categories remain underrepresented across the entire 2009–2024 timeframe. This pattern indicates that, although ML models are effective in exploiting temporal and environmental signals, structural determinants such as dwelling characteristics, land use, and network configuration are frequently excluded from predictive frameworks. The limited adoption of these predictors is commonly attributed to restricted access to household-level infrastructure data, fragmented data ownership, and privacy regulations.

The predictor landscape identified for RQ2 indicates a strong methodological emphasis on variables that are readily available through smart metering and environmental monitoring

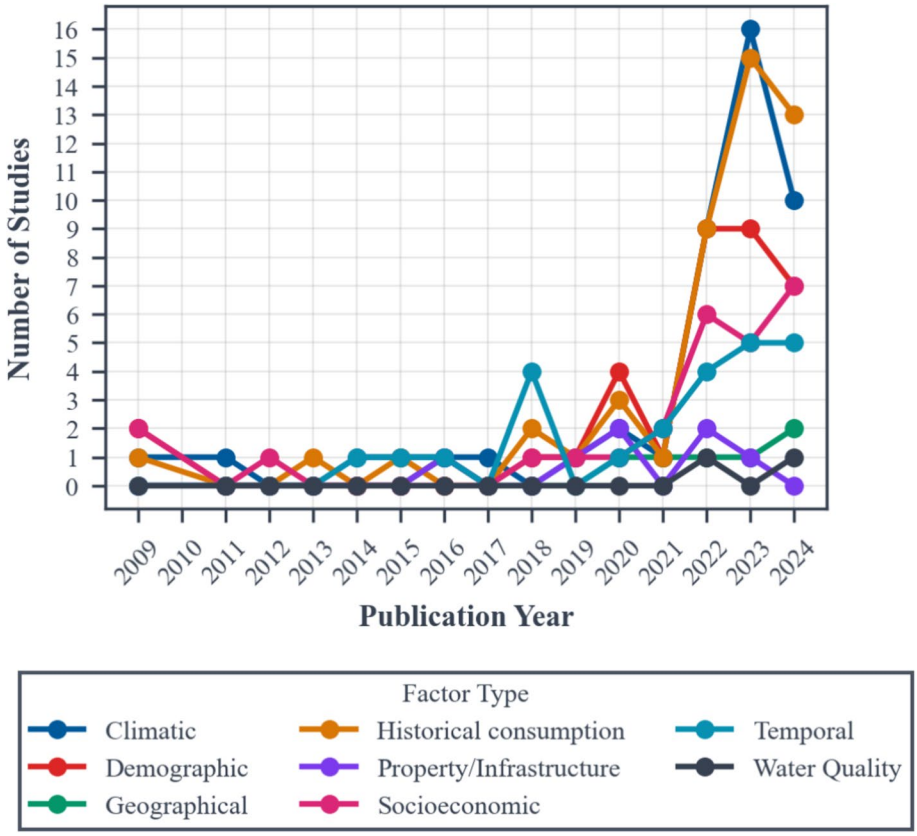

Fig. 7 Temporal distribution of predictor categories. Trends in usage of different types of predictors over time in ML models for water consumption forecasting

systems. While this emphasis supports short-term forecasting performance, it may restrict spatial transferability and reduce the capacity of models to inform long-term planning and policy-oriented analyses. These observations motivate the need for more integrated predictor strategies in future research on residential water consumption forecasting.

3.5 Performance Metrics Used to Evaluate Machine Learning Techniques for Household Water Consumption Prediction (RQ3)

Regarding RQ3, the performance metrics employed to evaluate machine learning models for residential water consumption forecasting were systematically identified and analyzed. Across the 80 reviewed studies, a total of 39 distinct evaluation metrics were reported, reflecting substantial heterogeneity in assessment practices. Despite this diversity, metric usage was found to be strongly concentrated around a limited subset of measures.

Applying a Pareto-based analysis, six metrics collectively accounted for approximately 80% of all reported mentions. These dominant metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), the coefficient of determination (R^2), Mean Squared Error (MSE), and Nash–Sutcliffe Efficiency

(NSE). The relative frequency of these metrics across the reviewed studies is summarized in Fig. 8. Among them, RMSE emerged as the most frequently adopted metric, followed by MAE and MAPE, indicating a strong preference for magnitude-based error measures in residential water demand forecasting.

The distribution of evaluation metrics across different model families is illustrated in Fig. 9. Neural network-based models, including recurrent and deep learning architectures, were predominantly evaluated using RMSE and MAE, reflecting their common application to continuous time series prediction tasks. Tree-based and ensemble methods exhibited more heterogeneous evaluation practices, incorporating both absolute error metrics and goodness-of-fit measures such as R^2 and NSE. Kernel-based models and linear baselines were similarly evaluated using combinations of scale-dependent and scale-independent metrics, depending on study objectives and data characteristics.

In addition to metric prevalence, reporting practices revealed variability in terminology and metric definition across studies. In particular, the term *accuracy* was frequently used in an ambiguous manner, often without explicit definition or distinction from regression-

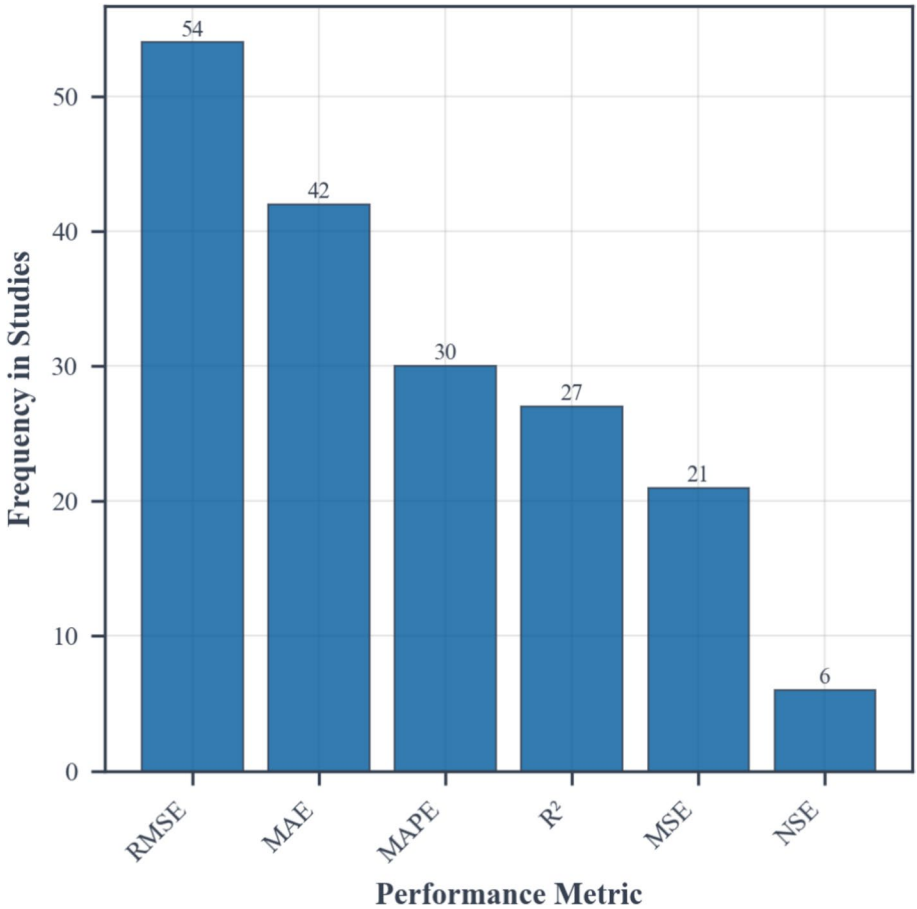

Fig. 8 Frequency of evaluation metric usage. RMSE is the most commonly used metric across reviewed studies

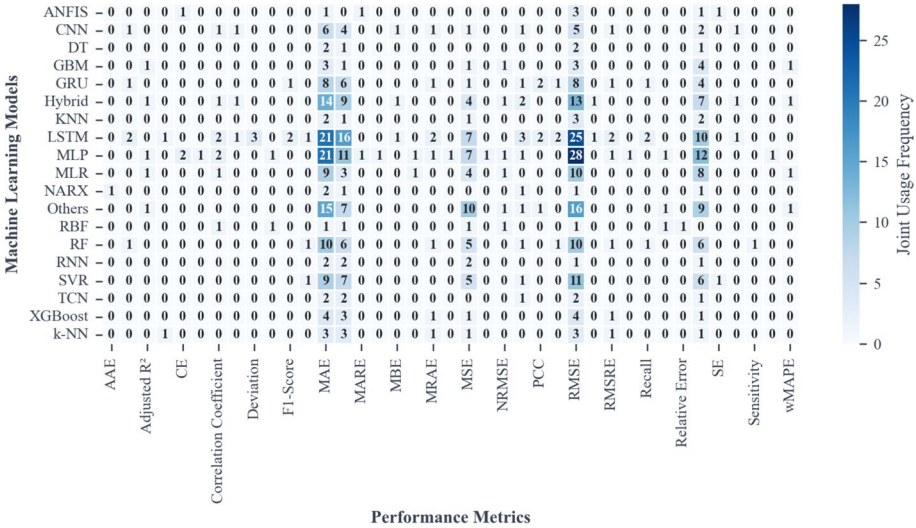

Fig. 9 Heatmap showing frequency of use of evaluation metrics across ML models. RMSE and MAE dominate, particularly for LSTM and MLP

specific performance measures. In several cases, *accuracy* was employed to refer to inverse error measures, normalized errors, or coefficients of determination, complicating direct comparison across studies. This lack of consistency in metric reporting underscores the fragmented nature of evaluation practices within the literature.

The findings for RQ3 indicate that evaluation strategies in residential water consumption forecasting are centered on a narrow set of conventional error metrics, with limited standardization in terminology and reporting. While this concentration facilitates comparison within subsets of studies, it also highlights the need for clearer metric definitions and more consistent evaluation frameworks in future research.

3.6 Expected Challenges in Predicting Household Water Consumption (RQ4)

To answer RQ4, we extracted and categorized the challenges reported in the primary studies using the same procedure applied in RQ3. Thirteen types of challenges were identified, as shown in Figs. 10 and 11.

The most common issue, reported in 29 studies out of 80 studies (e.g., [P04], [P17], [P19], [P24], [P26], [P35], [P63]), concerns the difficulty of identifying and interpreting relevant predictors. For example, [P17] addressed the integration of climatic features into IoT-based models, [P19] explored complementary variables to improve generalization, and [P24] incorporated new variables to enhance behavioral insights.

Accuracy enhancement, cited in 16 out of 80 studies (e.g., [P08], [P22], [P30], [P51], [P58]), was the second most prevalent challenge. Approaches included metaheuristic optimization ([P08]) and feature influence analysis ([P57]).

Data collection limitations, noted in 14 out of 80 studies ([P18], [P33], [P42], [P48], [P53], [P66]), involved the scarcity or incompleteness of training datasets. [P35] and [P48]

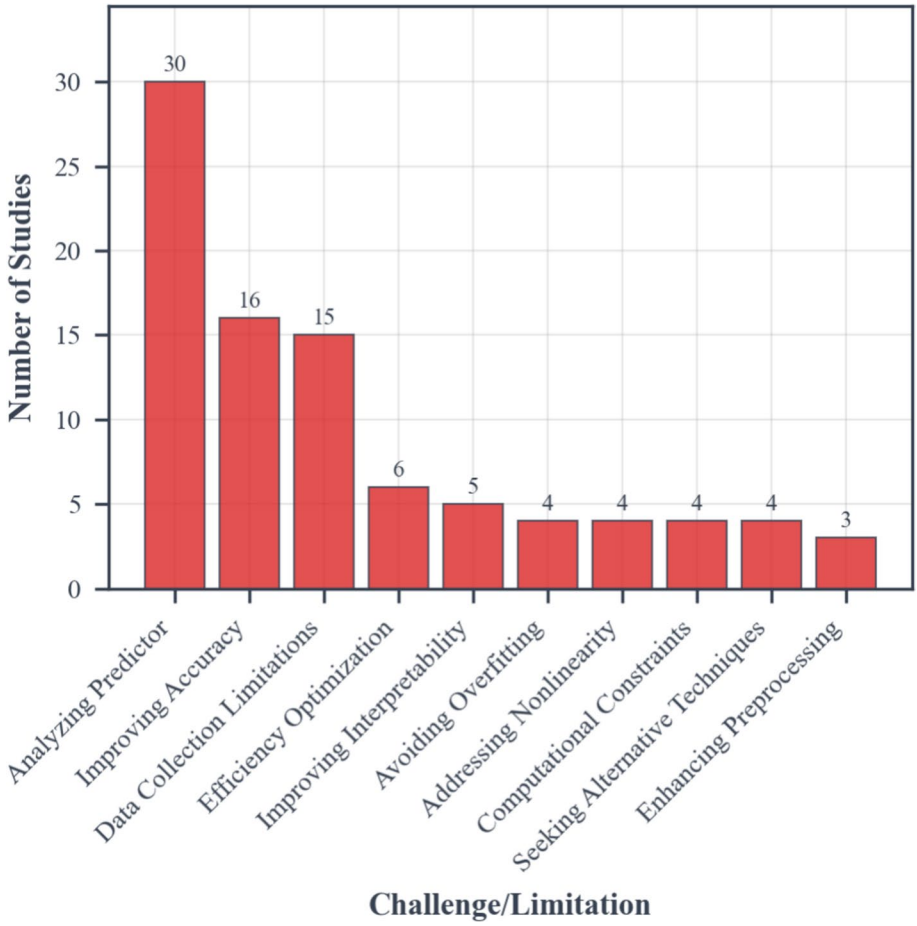

Fig. 10 Reported challenges in ML-based forecasting. Frequency of challenges discussed in the reviewed literature regarding ML for water usage prediction

highlighted the need for more representative and diverse data to avoid overfitting and improve reliability.

Other challenges, though less frequent, were still noteworthy: - **Efficiency optimization** ([P03], [P27], [P58])—reducing complexity without sacrificing performance. - **Interpretability** ([P18], [P29], [P52])—making model outputs more transparent. - **Preprocessing** ([P03], [P06])—addressing data quality prior to modeling. - **Nonlinearity** ([P35], [P44])—capturing nonlinear dynamics in traditional models. - **Model overfitting** ([P18], [P50])—ensuring generalizability. - **Alternative techniques** ([P06], [P45])—testing hybrid or novel ML methods. - **Computational constraints** ([P30], [P38])—hardware limitations. - **Adaptability** ([P02])—transferring models to new domains. - **Model evaluation** and **long-term forecasting** ([P29])—choosing metrics and extending horizons. - **Non-stationarity** ([P44])—handling temporal changes in data distribution.

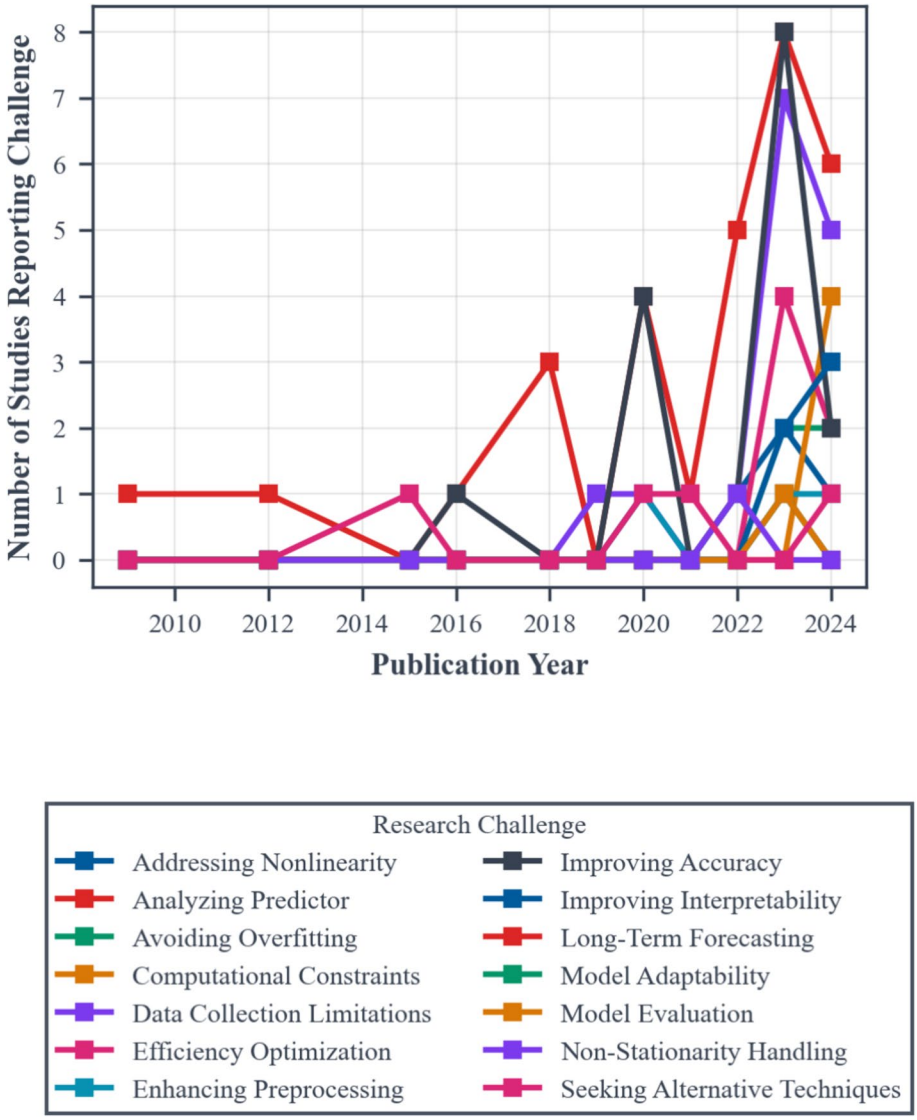

Fig. 11 Annual evolution of challenge mentions. Yearly frequency of cited challenges in household water consumption forecasting using ML models

Figure 10 shows that predictor analysis, accuracy improvement, and data availability were the dominant concerns. These directly impact the design and reliability of ML models for forecasting water usage.

Figure 11 highlights temporal patterns. Since 2022, challenges related to predictor analysis and accuracy have risen in tandem, suggesting a growing recognition of their interdependence. Moreover, data acquisition became a key concern in 2023, reflecting increasing awareness of the foundational role of high-quality data in ML performance.

The findings emphasize that enhancing accuracy, improving feature selection, and securing reliable data sources remain central challenges in the use of ML for water consumption forecasting.

4 Discussion

As part of the data extraction related to RQ1, we identified the most frequently prioritized characteristics of machine-learning models for predicting household water demand. Importance was defined by the frequency with which a characteristic was mentioned across studies. Among all characteristics, non-linear modeling capability emerged as the most prominent, maintaining high relevance throughout the period covered by this SLR. As noted in [P12], water demand is inherently non-linear and depends on multiple interacting factors. Climatic variables, such as temperature and precipitation, exhibit strong seasonal and short-term variability, and their interaction with socio-economic indicators further increases modeling complexity, thereby motivating the widespread adoption of non-linear models.

Models with strong non-linear capacity dominate the reviewed literature, particularly MLP, LSTM, and Random Forest. MLP remains popular due to its general applicability and simplicity, provided sufficient feature engineering is performed. However, its lack of an integrated mechanism for capturing temporal dependencies often limits its effectiveness for time-series demand forecasting. LSTM architectures address this limitation by explicitly modeling long-term temporal dependencies and are therefore well suited to residential water-demand data, albeit at the cost of increased data requirements and parameter tuning. Random Forest models offer a complementary alternative, combining non-linearity, robustness to overfitting, and improved interpretability, especially when static or non-temporal predictors are used. Hybrid models, including combinations of deep learning and traditional techniques (e.g., CNN+LSTM or MLP+Wavelet), can achieve high predictive accuracy but typically increase computational cost and reduce interpretability.

Model interpretability remains a notable gap across the reviewed studies. Although neural architectures (M1) frequently outperform other model families, they largely operate as black boxes, limiting transparency, accountability, and trust for utility managers and end-users. Recent advances in Explainable AI provide promising solutions to this challenge. Model-agnostic techniques such as SHAP enable the quantification of predictor effects in feed-forward models, while attention mechanisms in recurrent architectures can highlight influential temporal patterns [P63]. Nevertheless, explicit adoption of such explainability frameworks remains rare, with only a small number of recent studies incorporating them (Infant et al. 2025; Makumbura et al. 2024).

From an operational perspective, particularly in IoT-based smart metering systems, model complexity introduces additional constraints. Deep learning models often require computational and memory resources that exceed the capabilities of low-power microcontroller units deployed at the edge. In contrast, tree-based (M2) and linear (M4) models generally demand fewer resources and lower energy consumption, making them more suitable for edge deployment. Consequently, model selection should consider not only predictive accuracy but also computational efficiency and deployment feasibility.

Regarding RQ2, predictor selection is dominated by historical consumption, climatic variables, and calendrical features. The importance of climatic predictors has been con-

sistently highlighted in several studies [P05, P18, P21, P23, P41, P67], particularly in the context of climate change. Historical consumption data is widely used due to its strong autocorrelation and ability to capture trends and seasonal patterns [P28, P35]. However, reliance on historical data alone may lead to reduced accuracy during structural or behavioral changes [P07, P36], emphasizing the need for additional predictors.

Socio-demographic and property-related variables, such as income, household size, dwelling type, and land use, are fundamental for understanding residential water-use behavior [P15, P21, P28]. Despite their relevance, these predictors are included in only a limited number of studies, largely due to privacy constraints, data availability, and misalignment between socio-economic datasets and high-frequency smart meter data. Similarly, spatial predictors remain rare because utility datasets often lack linked GIS information or consistent location identifiers. As a result, most models rely primarily on temporal and meteorological inputs, limiting their applicability for long-term planning and policy analysis.

Regarding RQ3, evaluation practices exhibit substantial heterogeneity. Although 34 distinct performance metrics were identified, only a small subset accounts for more than 80% of total usage. RMSE and MAE dominate due to their simplicity and interpretability. However, inconsistent definitions, particularly of the term "accuracy", undermine reproducibility and cross-study comparison. For instance, accuracy has been used interchangeably with MAE [P19], classification-based measures [P38, P49], or R^2 [P72]. Such inconsistencies highlight the need for clearer reporting standards and alignment with established metric definitions.

Finally, addressing RQ4, the most frequently reported challenge concerns uncertainty in predictor selection, identified in 29 of the reviewed studies. This issue is compounded by data-quality problems, including missing values, sensor outages, and inconsistent temporal granularity [P17, P61]. While some studies apply interpolation or imputation techniques [P03, P55, P68], many rely on discarding incomplete observations, increasing the risk of bias [P61]. Additional challenges include inconsistent preprocessing practices, lack of standardized benchmarks, and growing concerns related to privacy and data security. The increasing adoption of fine-grained smart-meter data raises privacy risks, reinforcing the need for lightweight, privacy-compliant models deployed in decentralized Edge-IoT architectures rather than centralized cloud-based systems.

5 Synthesis and Practitioner Guide

To ensure the guidance below is evidence-based, we contextualize our findings using influence patterns identified through a citation analysis (full bibliometric data is provided in the supplementary repository). First, we map systematic associations between *model families* and *predictor categories* with a co-occurrence heat map (the "Model–Predictor Crosswalk"). Second, we operationalize these patterns into a prescriptive decision tree for model selection under common data conditions (granularity, horizon, exogenous availability, sample size, and constraints). Together, these artifacts provide structure beyond a narrative review, exposing regularities across studies, flagging underexplored intersections, and translating them into practical recommendations derived from our taxonomy.

5.1 Model–Predictor Crosswalk

Figure 12 displays how often each *model family* (rows M1–M6) co-occurs with each *predictor category* (columns P1–P7) across the primary studies. Cell color encodes normalized co-occurrence intensity, with darker shades indicating more frequent pairings.

The heat map is constructed in four steps. First, for each study we extract the set of model families employed (e.g., M1, M2, M6) and the set of predictor categories used (e.g., P2, P1, P3). Second, we form all *pairwise* co-occurrences (M, P) present in that study and increment the corresponding cell by one.[1] Third, we aggregate across studies to obtain a 6×7 count matrix. Fourth, we normalize counts to intensities (default: global normalization to $[0, 100]$). We also compute row- and column-normalized variants to examine within-row and within-column preferences.

Visual inspection of Fig. 12 reveals a distinct structural segmentation in the literature, evidenced by high-intensity clusters (darker cells) and sparse regions. We identify three dominant "attractor basins" that define current modeling practices:

(a) **The Sequence–Exogenous Cluster (M1 with P1, P2, P3):** The dark vertical block under M1 indicates that neural networks (specifically LSTM and GRU) are the standard choice when high-frequency historical data (P2) is combined with external drivers like weather (P1) and calendar events (P3). This strong association reflects the architectural advantage of recurrent units in fusing dense temporal history with exogenous covariates to capture volatility in hourly or day-ahead tasks.
(b) **The Tabular–Heterogeneous Cluster (M2 with P2, P6):** A secondary intensity cluster connects tree-based ensembles (M2) with property and infrastructure predictors (P6). Unlike neural networks, which dominate purely sequential tasks, tree models (Random Forest, XGBoost) are preferred in studies that rely on static or slowly changing household attributes (e.g., dwelling type, garden size), likely due to their native ability to handle mixed data types and interactions without extensive preprocessing.
(c) **The Baseline Cluster (M4 with P2):** The moderate intensity linking linear baselines (M4) with historical lags (P2) highlights their enduring role as parsimonious benchmarks, particularly in small-*N* or interpretability-driven contexts.

Conversely, the pale or empty cells highlight underexplored intersections. The limited co-occurrence of M1/M6 with socio-economic (P4) and economic (P5) predictors suggests a "blind spot" in the current deep learning literature: sophisticated architectures are rarely applied to long-horizon policy questions where these structural variables are most predictive. Where hybrids (M6) are used, they most commonly combine decomposition or statistical components with ML on residuals, again leveraging P2 and often P1/P3.

To validate that the identified clusters represent widespread community consensus rather than artifacts of aggregation, we conducted a sensitivity analysis comparing *study-level* versus *experiment-level* counting.

- **Aggregation Method:** Our primary heat map (Fig. 12) uses binary study-level counts (max 1 count per model family per paper) to prevent bias from single studies reporting

[1] If a study reports multiple experiments with the same (M,P), we count it once at the study level to avoid overweighting prolific papers; results are robust to counting at the experiment level.

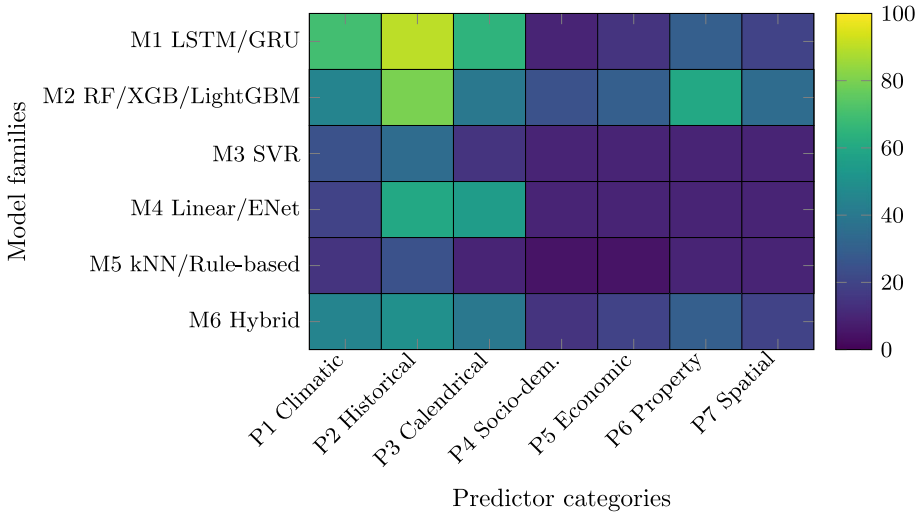

Fig. 12 Model–Predictor crosswalk (normalized co-occurrence intensities)

dozens of model variations.

- **Sensitivity Check:** We re-computed the matrix using cumulative experiment counts. The analysis revealed that while the absolute intensity of the M1–P2 cell increased (reflecting the tendency of deep learning studies to report extensive hyperparameter search grids), the **relative rank-ordering** of the top three "attractor basins" remained invariant. The Sequence–Exogenous cluster (M1) and Tabular–Heterogeneous cluster (M2) remained the two most dominant pairings under both counting schemes.
- **Conclusion:** This invariance confirms that the reported associations are robust to the unit of analysis and reflect genuine methodological preferences in the field.

These regularities inform the prescriptive guidance developed in Fig. 13. To assess robustness, we verified that the main patterns persist under row and column normalizations; we also recomputed the matrix by task context (forecast horizon and data granularity) and compared study-level with experiment-level counting. These checks did not alter the leading associations, though experiment-level counting slightly amplifies cells from highly comparative papers.

To ensure that these identified clusters represent widespread community consensus rather than artifacts of aggregation, we conducted a sensitivity analysis comparing our primary study-level counts against experiment-level counts. This check confirmed that, although experiment-level counting amplifies the intensity of deep learning cells, due to the tendency of such studies to report extensive hyperparameter search grids, the relative rank-ordering of the dominant clusters remains invariant. Specifically, the M1 and M2 basins persist as the leading pairings under both schemes, validating that the structural segmentation of the literature is robust to the unit of analysis.

These regularities inform the prescriptive guidance developed in Fig. 13. However, the heat map reflects *use*, not necessarily *performance*. Heterogeneous objectives, horizons, and metric choices limit causal interpretation, and reporting bias or variable availability

can skew co-occurrences. We therefore treat the crosswalk as evidence-informed context for model selection rather than a performance ranking. When applying these insights, we recommend reporting at least one scale-dependent error (RMSE or MAE) and one scale-free metric (e.g., sMAPE or MASE) to maintain comparability across households and settings.

5.2 Decision Tree for Model Selection

Figure 13 summarizes a prescriptive decision process derived from the heat map and the taxonomy in Section 5.1. The tree first splits by data granularity, high-frequency series

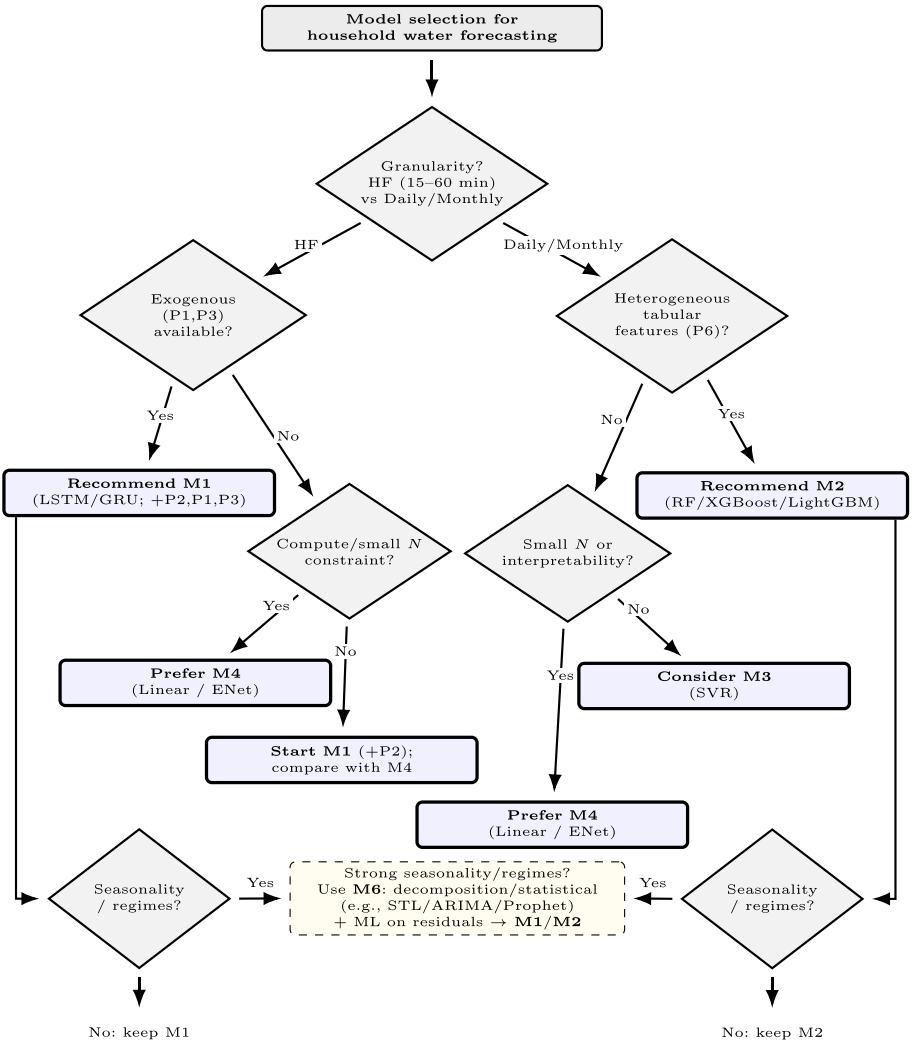

Fig. 13 Decision tree for model selection: M1 for high frequency data with exogenous signals (P1, P3), M2 for heterogeneous tabular contexts with property or infrastructure descriptors (P6), M4 when small N or interpretability is binding, M6 when seasonality or regime structure is pronounced

(≈ 15–$60\,\mathrm{min}$) versus daily/monthly aggregates—because sequence models benefit most from dense temporal signals, whereas aggregated settings behave more like heterogeneous tabular prediction. It then branches on exogenous drivers (P1/P3) and on heterogeneous property/infrastructure descriptors (P6), and finally on practical constraints (sample size and interpretability).

For high-frequency data with P1/P3 available alongside target lags (P2), we recommend **M1** (LSTM/GRU; 1D CNN when used), which captures short- and long-range dependencies. When computation or sample size is restrictive, or when strict interpretability is required, prefer **M4** (Linear/Elastic Net) as a compact baseline and benchmark against M1 to calibrate gains versus complexity.

For daily or monthly settings where P6 is informative, **M2** (tree ensembles: RF, XGBoost/LightGBM) is typically favored for its robustness to mixed data types, interactions, and missingness. If P6 is weak or absent, choose by constraints: with small N or interpretability needs, prefer **M4** (Linear/ENet); otherwise, when modest nonlinearity is suspected but the feature space remains limited, consider **M3** (SVR).

Across both branches, when the series exhibits strong seasonality or regime structure, adopt **M6** (hybrids) that combine decomposition or statistical components with ML on residuals (e.g., STL/ARIMA/Prophet $\rightarrow$ M1/M2); fuzzy components, when present, are used only within such hybrids.

To assess the robustness of this heuristic guidance, we conducted a citation analysis to contextualize the prominence of specific research streams, identifying two dominant 'anchor' strands: M1 (Neural Networks) paired with P1, P2, and P3 predictors, and M2 (Tree Ensembles) paired with P6 attributes. While the full bibliometric results (Total Citations and CPY) are documented in the supplementary repository, this validation confirms that the tree's terminal nodes align with historically influential workflows. Specifically, the branch prioritizing sequence models (M1) for high-frequency exogenous tasks is validated by foundational works such as [P04] and [P06], while the branch favoring tree ensembles (M2) for heterogeneous tabular data captures key studies like [P08] and [P15]. This alignment suggests that the decision tree not only reflects widespread community consensus but also steers practitioners toward methodological choices that have yielded high scientific impact.

To make the guidance more concrete, consider a few representative scenarios that trace typical paths through Fig. 13. In a utility that logs subdaily demand ($\approx$15 to 60 min) and also has weather and calendar variables P1/P3 together with target lags P2, the evidence points to M1 (LSTM/GRU, 1D CNN when used). Sequence models leverage dense temporal structure and exogenous signals; a compact M4 (Linear/ENet) baseline can be run in parallel to quantify the incremental gain.

If the setting is still high frequency, exogenous drivers are unavailable, and the sample or compute budget is tight (small N, on-device inference), a parsimonious M4 is preferable, with a small M1 as an optional comparator. Conversely, when exogenous drivers are absent but there is abundant history (large N), starting with M1 and benchmarking against M4 is sensible, as long-range temporal structure can be learned from P2 alone.

When data are daily or monthly and rich property or infrastructure descriptors are available (P6 alongside P2, optionally P1/P3), the prevailing choice is M2 (tree ensembles: RF, XGBoost, LightGBM). These methods handle mixed types, interactions, and missingness robustly. If P6 is weak or unavailable and either small N or interpretability is a priority, M4

is usually the safer option. If modest nonlinearity is suspected and the feature space remains limited, M3 (SVR) is a competitive alternative and should be compared to M4.

Across both branches, if the series exhibits strong seasonality or regime structure, the path diverts to M6 (hybrids): first impose structure via decomposition or a statistical component (e.g., STL, ARIMA, Prophet), then learn residuals with ML, typically M1 when operating at high frequency or M2 in tabular contexts. This "structure-then-learn" approach is especially effective for long horizons and policy scenarios.

The decision tree encodes patterns of use rather than an absolute ranking. We recommend validating the selected path via nested cross-validation, reporting at least one scale-dependent error (RMSE or MAE) and one scale-free metric (e.g., sMAPE or MASE). Furthermore, for longer-horizon planning, it is advisable to stress-test sensitivity to the inclusion of socio-demographic and economic signals (P4/P5) to maintain comparability across households and diverse contexts.

6 Related Work

In recent years, the application of machine learning (ML) to water resource management has gained increasing attention, particularly in the forecasting of household water consumption. Several review studies have been published to synthesize current trends, identify methodological advancements, and highlight key limitations in this field.

Sit et al. (2021) provide a broad review of ML applications across diverse domains of water management, including flow forecasting and distribution systems. Their study highlights the effectiveness of LSTM networks and hybrid models but does not specifically address the challenges of forecasting household water use. Adamowski and Chan (2015) offer a more focused perspective, reviewing soft computing techniques applied to water demand forecasting between 2005 and 2015. They note that neural networks often outperform other methods, while also emphasizing the difficulty of selecting a universally superior model. Given the rapid evolution of ML techniques, however, their review is now dated and requires updating. Shen (2018) shift the focus toward deep learning (DL), discussing its broader potential in the water sciences.

Although they underscore DL's capacity to enhance process efficiency and scientific insight, they do not explicitly examine applications in household water forecasting. Zubaidi et al. (2020) explore ML and data analytics in the context of digital water metering, identifying demand forecasting as one of five key application areas. Nevertheless, their discussion of consumption prediction remains brief and technically limited. Similarly, Herath and Mittal (2022) examine AI in smart cities, particularly after COVID-19. While water management is included, the treatment is broad and lacks methodological rigor. A comparative overview of these related works is presented in Table 1.

Other studies contribute more indirectly. Fu et al. (2022) review the role of DL in urban water management, pointing to early applications in consumption prediction, leakage detection, and contamination analysis. However, they stop short of evaluating household-specific ML models in depth. Mosavi et al. (2018) investigate hybrid AI and optimization techniques for streamflow prediction. Although methodologically insightful, their work is oriented toward natural systems rather than residential contexts. Garzón et al. (2022) review ML-based surrogate models for simulating urban water networks, useful for optimization but

Table 1 Comparison of related review articles

Reference	Scope	Household Consumption	ML Models and Predictors Discussed	Main Contributions / Gaps
Sit et al. (2021)	ML in water resources management	Partial (general water domains)	LSTM, hybrid models; general predictors	Demonstrates ML success; lacks focus on household prediction
Adamowski and Chan (2015)	Soft computing for water demand	Yes (2005–2015 focus)	NN, fuzzy models, metaheuristics	Strong baseline; needs update to reflect recent developments
Shen (2018)	DL in water sciences	No	High-level DL overview; no specific models for demand	Valuable for DL motivation; no household-level insights
Zubaidi et al. (2020)	Digital water metering	Briefly mentioned	ML + analytics; categorized 5 applications	Demand prediction included, but with minimal technical depth
Herath and Mittal (2022)	AI in smart cities	Very limited	General AI technologies post-COVID	High-level perspective; not technical nor predictive
Fu et al. (2022)	DL in urban water management	Partial	CNN, LSTM; water quality, leakage, usage	Highlights early-stage DL use; lacks model-level evaluation
Mosavi et al. (2018)	AI + optimization for streamflow	No	NN, SVM, hybridized with GA, PSO	Relevant to methodology, not to household consumption
Garzón et al. (2022)	Surrogate ML models in water networks	No	ML-based surrogates for network simulation	Useful for optimization, not specific to demand forecasting
Gonzalez-Cebollada et al. (2020)	Urban water demand modeling	Partial (urban context)	GIS, agent-based models, dynamic systems	Strong conceptual framing; not focused on ML models
This work	**ML for household water forecasting**	**Yes (household-level SLR)**	**MLP, LSTM, RF; 7 predictor types; 34 evaluation metrics**	**Most complete SLR; identifies key models, predictors, metrics, and open challenges**

Bold entries indicate the characteristics and contributions of the present work, emphasizing its focus on household-level consumption, the range of ML models and predictors analyzed, and the comprehensive nature of the SLR

not designed for direct consumption forecasting. Gonzalez-Cebollada et al. (2020) provide a valuable conceptual review of urban water demand modeling, advocating for transdisciplinary approaches that integrate socioeconomic, environmental, and behavioral dimensions. Yet their emphasis lies on conceptual integration rather than ML-driven prediction.

In contrast, the present study delivers a focused and comprehensive systematic literature review (SLR) dedicated to ML techniques for household water consumption forecasting. Based on 80 primary studies published between 2009 and 2024, our review introduces a structured taxonomy of the most widely used ML models (e.g., MLP, LSTM, Random Forest), seven categories of predictors, and 34 evaluation metrics. Furthermore, it synthesizes the main benefits, challenges, and open research directions. By offering a technically detailed and domain-specific synthesis, this work addresses a critical gap and provides guidance for both future research and practical implementation.

7 Threats to Validity

7.1 Internal Validity

Screening and coding errors may affect which studies were included, and how they were labeled. To reduce this risk, two reviewers independently applied the inclusion and exclusion criteria, piloted and refined a codebook, and resolved disagreements through discussion, and when needed, third reviewer adjudication. The taxonomy was frozen before quantitative synthesis in order to reduce outcome driven drift. Residual risks remain, for example borderline cases with ambiguous hybrids, coder drift over time, or misinterpretation of pipelines that are insufficiently documented. Our sensitivity checks, which include counting co occurrences at the study level and at the experiment level, and re computing normalizations, mitigate these risks, although they do not eliminate them. Furthermore, the quality assessment stage (Step 5) relies on manual scoring of dimensions such as credibility and relevance. Despite our collaborative consensus-based resolution process, this step retains an element of subjectivity inherent to expert-based evaluation, which may influence the final quality scores.

7.2 Construct Validity

Construct validity concerns whether our operationalizations accurately reflect the intended concepts. First, the model families M1–M6 represent usage-based categories rather than theoretical taxonomies. Gaussian processes, ARIMA or Prophet models, and fuzzy systems were counted only when used as components of hybrid approaches (M6). This exclusion of standalone classical statistical models, unless hybridized, introduces a limitation regarding the comprehensiveness of the modelling landscape covered in this review. Although the decision was necessary to maintain a manageable scope focused on machine-learning-specific modelling characteristics, it filters out studies where statistical models operate as primary forecasters rather than benchmarks or auxiliary components.

As a result, the prominence of ML families (M1–M6) reported here should be interpreted as a distribution within the machine learning domain, not as a representation of all modelling approaches used in household water forecasting. Second, predictors P1–P7 are coded by their semantic origin. In ambiguous cases, pure target lags default to P2, which may obscure alternative interpretations such as engineered seasonality features. Third, evaluation metrics are grouped according to predominant usage patterns. Choices such as MAPE or sMAPE versus MASE can lead to different comparative rankings, particularly in the presence of zeros or heavy-tailed errors. Fourth, the model–predictor heat map encodes frequency of co-occurrence rather than effect size or causal contribution.

Its results may be sensitive to normalization schemes and to the rule that each pair (M, P) is counted once per study. Finally, regarding the prescriptive decision tree, although its paths were validated against high-impact "anchor" studies and sensitivity checks, the framework remains a descriptive heuristic based on community consensus. Consequently, it reflects widespread usage patterns, potentially subject to popularity bias, rather than a normative benchmark derived from controlled experimentation.

7.3 External Validity

Generalizability is limited by the composition of the corpus. Most primary studies involve specific climates, metering technologies such as AMR or AMI, regulatory contexts, and data granularities. A further significant threat concerns the limited use of socio-economic and spatial predictors in the included studies. Variables such as household demographics, tariff structures, and land use patterns are known to influence medium- and long-term consumption behaviour, yet they appear only sporadically in the ML literature due to privacy restrictions and data availability constraints. This scarcity biases the synthesized patterns toward short-term forecasting scenarios driven principally by historical and climatic data. Consequently, transferring conclusions across regions, horizons, or infrastructure settings, particularly where structural drivers are prominent, may require re-validation. Additionally, two forms of bias limit the scope of our findings. First, the exclusion of non-english sources introduces a language bias, potentially overlooking regionally significant studies published in local journals, particularly from non-anglophone regions facing acute water stress. Second, we acknowledge the risk of publication bias (the "file-drawer problem") common in applied machine learning: studies reporting state-of-the-art results using complex architectures like LSTMs are more likely to be published than those reporting negative results or negligible improvements over simpler baselines. This may result in an overestimation of the practical superiority of complex deep learning models relative to parsimonious alternatives. Finally, the time window, 2009 to 2024, may under-represent very recent developments. In practice, model selection should be reassessed under local constraints, including data availability, feature sets, compute budgets, and policy interpretability, and validated with blocked or rolling-origin cross-validation.

7.4 Conclusion Validity

We did not conduct a statistical meta-analysis of effect sizes, because objectives, horizons, feature sets, and metrics are heterogeneous. Consequently, the heat map and the decision tree should not be interpreted as tests of performance differences. Multiple visual inspections and comparisons across cells introduce a risk of over-interpretation. The citation analysis adds context for influence, totals and citations per year, but it is subject to the coverage of the indexing source and to the retrieval date. High citation counts do not imply superior accuracy. We therefore treat our synthesis as descriptive evidence that can guide model choice, not as confirmatory inference. Furthermore, the relevance score of 58.3% reported in the quality assessment reflects the nascent nature of a significant portion of the corpus (2020–2024). While this ensures coverage of state-of-the-art deep learning techniques, it implies that our synthesis relies heavily on emerging evidence whose long-term impact and methodological consensus have not yet been fully established by the community.

7.5 Reproducibility and Mitigation

We specify inclusion and exclusion rules, provide operational definitions for taxonomy labels, and document the aggregation steps behind the heat map and the citation analysis, including the CPY formula and the normalization variants. Where possible, we make the extraction sheet available, and we recommend releasing scripts to regenerate the figures.

For applied use, we advise reporting at least one scale dependent error, RMSE or MAE, and one scale free metric, for example sMAPE or MASE, performing ablations on exogenous predictors, P1 and P3, and on property or infrastructure descriptors, P6, and validating selections with nested cross validation or with rolling origin cross validation.

8 Conclusions and Research Opportunities

The findings from 80 primary studies were synthesized to examine the application of Machine Learning (ML) techniques for household water consumption forecasting. Model characteristics, predictor usage, performance metrics, and reported challenges were analyzed under the guidance of four research questions.

Regarding model characteristics (RQ1), a consistent emphasis was identified on the ability to capture nonlinear relationships, reflecting the multifactorial nature of residential water use. Models such as Multilayer Perceptron (MLP), Long Short Term Memory (LSTM) networks, and Random Forest (RF) were widely adopted across the literature. However, a critical research gap was observed with respect to model transparency. Interpretability techniques, including SHAP and attention mechanisms, were rarely incorporated, which may limit the applicability of ML models in policy oriented contexts where accountability and explainability are required.

Concerning modeling implications (RQ2), climatic variables, historical consumption patterns, and demographic features were identified as the most frequently employed predictors. In contrast, contextual variables such as socioeconomic status, geographic conditions, and property infrastructure were found to be underrepresented. This underutilization was primarily attributed to limitations in data availability and privacy related constraints. Evaluation practices (RQ3) were observed to rely predominantly on a narrow set of performance metrics, namely Root Mean Squared Error, Mean Absolute Error, Mean Absolute Percentage Error, and Mean Squared Error. These metrics dominated the literature due to their simplicity and ease of interpretation. Nevertheless, inconsistencies in metric definitions were detected, particularly in the use of the term *accuracy*, which was often applied ambiguously. This observation highlights the need for standardized reporting practices to support reproducibility and comparability across studies.

Finally, the most significant challenges identified (RQ4) were related to the selection and analysis of predictors capable of generalizing across heterogeneous regions and temporal scales. These challenges were closely associated with persistent data quality issues, including missing sensor values and inconsistent preprocessing procedures. It was concluded that addressing these methodological and data related limitations, particularly those affecting transparency and interpretability, is essential for enabling the transition of ML based forecasting approaches from experimental research settings to reliable operational deployment by water utilities.

8.1 Future Directions

Several research directions were identified as promising based on the gaps revealed through the proposed taxonomy, heat map, and decision tree analysis.

(R1) **Socioeconomic signals at extended horizons:** It should be examined whether predictors P4 and P5 improve forecasting performance at weekly and monthly horizons when combined with models M1 and M6.

(R2) **Spatial generalization:** Predictor P7 should be combined with domain adaptation or meta learning techniques in order to facilitate transferability across different neighborhoods.

(R3) **Interpretable temporal models:** Attribution mechanisms for model M1 under predictor groups P1 and P3, including linear attention and SHAP based methods, should be developed to reveal the underlying drivers of consumption dynamics.

(R4) **Standardized hybrid pipelines:** Structure then learn approaches, such as decomposition based methods followed by neural or ensemble models, should be systematically benchmarked. Specific ablation studies should be conducted to assess the contribution of decomposition strategies and residual learning.

(R5) **Data efficiency:** Model robustness in small sample regimes should be enhanced through the use of regularized M4 baselines and few shot variants of models M1 and M2.

(R6) **Benchmarking and reporting standards:** A public household level benchmark dataset with fixed training and testing splits, together with a reporting checklist, should be established to ensure reproducibility and enable consistent comparison across diverse consumption profiles.

(R7) **Resource efficient Edge AI:** Techniques such as model compression, quantization, and TinyML should be investigated to enable the deployment of forecasting models directly on resource constrained smart meters.

In conclusion, ML models were found to exhibit substantial potential for predicting residential water consumption, particularly when nonlinear dynamics and temporal dependencies were effectively captured through appropriate model architectures and feature representations. However, the translation of technical performance improvements into operational value requires forecasting systems to be transparent, interpretable, and auditable, while simultaneously addressing concerns related to privacy, equity, and governance. By consolidating evidence from 80 primary studies and introducing a structured taxonomy, a model predictor heat map, and a prescriptive decision tree, this review provides an evidence informed foundation for the development of reliable and trustworthy forecasting tools for the water sector.

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Declarations

Competing Interest The authors declare no competing interests.

Ethical Approval All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors," as found in the Instructions for Authors.

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